INCH POUND

MIL-DTL-50700A (OS)
18 October 2005
SUPERSEDING
MIL-I-50700 (MU)
23 December 1970

DETAIL SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, JAU-2/A25 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Initiator, Propellant Actuated, JAU-2/A25, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government Document.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550 Ammunition and Special Weapons, General Specification for

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Products Branch (Code E143), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-413 Visual Inspection Guide for Rubber O-Rings

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

MUNITIONS COMMAND

D11738406	Initiator, Propellant Actuated, JAU-2/A25, Metal Parts Assembly
D10551447	Initiator, Propellant Actuated, Metal Parts Assembly
C9252573	Carton, Packing, Ammunition for Initiator, M113
C9252574	Box, Packing, Ammunition for Initiator, M113

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Application for copies may be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head, Technical Information Products Branch (Code E143), Indian Head, MD 20640.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4 Sampling Procedures and Tables for Inspection by Attributes

ANSI/ASQ A8402 Quality Management and Quality Assurance - Vocabulary

(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B 46 1-2002 Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at http://www.asme.org or from the American Society of Mechanical Engineers, 22 Law Dr. P.O. Box 2900, Fairfield, NJ 07007-2900)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 pounds plus or minus 5 pounds is applied at the steady rate of 0.125 inches per minute.

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.4 Residual magnetism. The metal parts assemblies shall not deflect the indicator of the compass more than 5 degrees in either direction when passed in front of the compass at a distance of 5 inches.

3.5 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects. The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr. The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter. The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections.

a. First Article inspection (see 4.2)
b. Lot acceptance inspection (see 4.3)

4.1.1 Quality assurance terms and definitions. Reference shall be made to ANSI/ASQ A8402 to define the quality assurance terms used.

4.1.2 Inspection. Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample. At the beginning of regular production, the contractor shall submit an initial production sample to a government approved facility for evaluation. The sample shall consist of 50 units from each head treated lot for hardness tests, 50 shear pins from each lot, 23 units for protective finish tests, to be tested in accordance with Table I and 10 sets of unassembled parts for visual and dimensional examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

Table I. First article samples and tests.

Paragraphs		Test	Sample size	Type of test	
Requirement	Test			Destr	Non-Destr
3.2	4.4.1	Shear Pin Strength	50	X	—
3.3	4.4.2	Protective Finish	In quantities set up in applicable specification	X	—
Dwg.	4.4.3	Hardness	50	X	—
3.4	4.4.4	Residual Magnetism	each	—	X

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests. After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product. The product shall be submitted in accordance with ANSI/ASQ Z1.4.

4.3.1.2 Size of lot. A lot shall consist of an identifiable group of metal parts assemblies consisting of components produced by a homogeneous process.

4.3.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The surface roughness comparison specimens prescribed in ASME B46.1-2002 shall be used as a basis of comparison for surface roughness determination. Four power (4×) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II. Sample size and acceptance, individual defect.

Lot Size	Sample Size	Major		Major	
		Accept	Reject	Accept	Reject
1 — 300	100% Examination	—	—	—	—
301 — 1300	110	1	2	2	3
1301 — 3200	150	2	3	3	4
3201 — 8000	225	3	4	4	5
8001 — 22000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

Table III. Sampling size and acceptance, total defects in class.

Total Allowable Defects per Sample Size				
Sample Size for Major Defects No. of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Size for Minor Defects No. of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Pin Firing, Dwg. C10521573 Covering a detail of Dwg. D11738406

Characteristics	Defect	Method of Inspection
Critical		
1	Diameter of striker	Gage
2	Length of striker	Gage
3	Metal defective	Visual
Major		
101	Concentricity of “0” ring groove with body	Gage
102	Concentricity of striker with body	Gage
103	True position of shear pin hole with body	Gage
104	Diameter of “0” ring groove, min.	Gage
105	Diameter of body, min.	Gage
106	Diameter of shear pin hole, max.	Gage
107	Depth of shear pin hole, max.	Gage
108	Length of “0” ring groove	Gage
109	Location of shear pin hole	Gage
110	Permissible variation of “0” ring groove	Gage
111	Protective finish inadequate or defective	Visual
112	Burr or sharp corner, except where permitted	Visual
113	Foreign matter	Visual
Minor		
201	Diameter of "0" ring groove, max.	Cage
202	Diameter of body, max.	Gage
203	Diameter of shear pin hole, min.	Gage
204	Depth of shear pin hole, min.	Gage
205	Depth of counterbore	Gage
206	Total length	Gage
207	Length of body to “0” ring groove	Gage
208	Location of marking	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

209	Marking missing, incorrect or illegible	Visual
210	Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Housing, Dwg. D11731622, covering a detail of Dwg. D11738406

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
Major		
101	Pitch diameter of external thread, min.	Gage
102	Major diameter of external thread, min.	Gage
103	Pitch diameter of internal thread, max.	Gage
104	Minor diameter of internal thread, max.	Gage
105	Concentricity pitch diameter of external thread with pitch diameter of internal thread	Gage
106	Concentricity of center hole with center cavity	Gage
107	Concentricity of cavity at large end with internal thread	Gage
108	Concentricity of internal thread with center cavity	Gage
109	Concentricity of large body with pitch diameter of internal thread	Gage
110	Perpendicularity of base of middle cavity with wall of middle cavity	Gage
111	Perpendicularity of base cavity at large end with pitch diameter of internal thread	Gage
112	Symmetry of shear pin hole with center cavity	Gage
113	Symmetry across flats with internal thread	Gage
114	Diameter of shear pin shaft, max.	Gage
115	Diameter at top of shear pin hole	Gage
116	Diameter of center hole, max.	Gage
117	Diameter of counterbore, large end, max.	Gage
118	Diameter of shoulder	Gage
119	Depth of shear pin hole	Gage
120	Length of internal thread, min.	Gage
121	Length of internal thread, max.	Gage
122	Width of thread undercut	Gage
123	Location of top end of shear pin hole	Gage
124	Location of shear pin shaft	Gage
125	Protective finish inadequate or defective	Visual
126	Surface finish of center cavity, chamfer at mouth of large end and sidewalls of center cavity	Visual
127	Burr or sharp corner	Visual
128	Foreign matter	Visual
Minor		
201	Pitch diameter of internal thread, min.	Cage
202	Pitch diameter of external thread, max.	Gage
203	Diameter of center hole, min.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

204	Diameter of flange, small end	Gage
205	Diameter of center cavity	Gage
206	Diameter of cavity large end	Gage
207	Diameter of counterbore, large end, min.	Gage
208	Diameter of chamfer, large end	Gage
209	Diameter of body, large end	Gage
210	Diameter of shear pin shaft, min.	Gage
211	Diameter across flats	Gage
212	Depth to chamfer of end cavity	Gage
213	Depth to bottom of cavity, large end	Gage
214	Depth of counterbore at chamfer end	Gage
215	Total length	Gage
216	Length of flats	Gage
217	Length of hole between cavities	Gage
218	Thickness of flange	Gage
219	Thickness of shoulder	Gage
220	Location of internal thread, max.	Gage
221	Location of internal thread, min.	Gage
222	Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Pin, Shear, Dwg. B11731626, covering a detail of Dwg. D11738406

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
Major		
101	Diameter, min.	Gage
102	Total length, min.	Gage
103	Surface finish improper	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106.	Foreign matter	Visual
Minor		
201	Diameter, max.	Gage
202	Total length, max.	Gage

4.3.2.4 Bracket, Mounting, D11731741 covering a detail of Dwg. D11738406

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of internal thread, max.	Gage
102	Minor diameter of internal thread, max.	Gage
103	True position of boss with pitch diameter of internal thread	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

104	True position of mounting holes with pitch diameter of internal thread	Gage
105	Diameter of thread lead	Gage
106	Length of effective thread, min.	Gage
107	Length of thread, max.	Gage
108	Metal defective	Visual
109	Protective finish inadequate or defective	Visual
110	Burr	Visual
111	Foreign matter	Visual
Minor		
201	Pitch diameter of mounting holes	Gage
202	Diameter of mounting holes	Gage
203	Length across open end of bracket	Gage
204	Length across front end of bracket	Gage
205	Total length	Gage
206	Location of mounting holes	Gage
207	Location of large hole	Gage
208	Height of shoulder	Gage
209	Total height including boss	Cage
210	Surface finish improper	Visual

4.3.2.5 Ignition Device Retainer, Dwg. C11738408, covering a detail of Dwg. D11738406

Characteristics	**Defect**	**Method of Inspection**
Critical		
1	Diameter of ignition device pocket	Gage
2	Depth of ignition device pocket	Gage
3	Metal defective	Visual
Major		
101	Thickness of wall between ignition device pocket and relief cavity	Gage
102	Protective finish inadequate or defective	Visual
103	Surface finish of ignition device pocket, improper	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
MINOR		
201	Diameter across chamfer	Gage
202	Diameter of relief cavity	Gage
203	Diameter of body	Gage
204	Total length	Gage
205	Radius or chamfer missing or incorrect	Visual
206	Surface finish improper, except as otherwise classified	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.6 Assembly, initiator, propellant actuated, JAU-2/A25 - Metal Parts, Dwg. D1l738406

Characteristics	Defect	Method of Inspection
Critical		
1	Firing pin protrusion	Gage
2	Metal defective	Visual
Major		
101	Any part missing	Visual
102	Improper assembly	Visual
103	Thread damaged or not full	Visual
104	Burr	Visual
105	Foreign matter	Visual
Minor		
201	“O”ring lubricant missing or inadequate	Visual
202	Marking missing, illegible or incorrect	Visual
203	Varnish or marking missing or inadequate	Visual

4.3.2.7 Carton, packing, ammunition for initiator, Dwg. C9252573

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Gap between body and cover	Manual
202	Contents loose	Manual
203	Metal ends loose or distorted	Manual
204	Cut or damaged through all impregnated layers	Visual
205	Marking missing, incorrect or illegible	Visual

4.3.2.8 Box, packing, ammunition for initiator, Dwg. C9252574

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Hardware missing, loose, broken or improperly engaged	Manual and Visual
203	Strapping missing, loose, broken or improperly assembled	Manual and Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect or illegible	Visual
207	Number of units incorrect	Visual

4.3.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Shear pin strength. Fifty pins from each lot of shear pins tested by the contractor shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of 3.3 and the applicable specification shall be cause for rejection of the lot.

4.3.3.3 Hardness. Fifty units from each heat-treated lot of components shall be tested for the values noted on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.4 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirement of 3.4 shall be cause for rejection of the assembly.

4.4 Test methods and procedures.

4.4.1 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with 3.2. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.4.2 Protective finish. Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.3 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the applicable drawing.

4.4.4 Residual magnetism. The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The JAU-2/A25 initiator is used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The initiator was developed specifically for use in military aircraft, and has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

6.3 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.4 Contract packaging, packing marking. The following requirements should be specified in the contract.

6.4.1 Packing marking for Level B (Navy) should be in accordance with Drawings 9252573 and 9252574.

6.5 Subject term (key word) listing
aircrew emergency escape systems

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I041R1.

Custodian:
Navy - OS

Preparing activity
Navy- OS
(Project 1377-2005-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.